



Olfactory Dysfunction in Post-Acute COVID-19: clinical characterization, hedonic asymmetry, and machine-learning-based predictor identification

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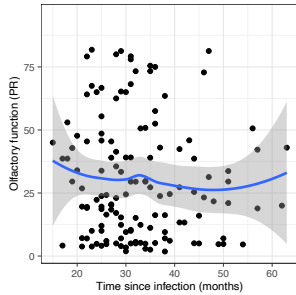
Background

- Olfactory dysfunction (OD) is an early and prominent symptom acute Sars-CoV2 infection and may persist for several years ^{1,2}
- Persistent smell loss affects millions of former SARS-CoV2 patients worldwide, with major quality of life implications ³
- Patients with Post-Acute Sequelae of COVID-19 (PASC) were tested in the Post-COVID Center Erlangen with Sniffin' Sticks TDI test and valence ratings of the identification test
- Access to cognitive, sociodemographic, and biomarker parameters from other tests ⁴

N = 144

Assessment

- 44.5%** of participants were tested for olfactory dysfunction, despite time since infection of 0.5 - 5.25 years
- Olfactory function (Age- and gender corrected, percentile rank ⁵) **deviated significantly from average** ($u = 50$; $W = 1385$; $P < .001$; $r = 0.64$)
- Olfactory function (Age- and gender corrected, percentile rank) did **not correlate with longer time since infection** ($\rho = -0.07$; $P = .43$)

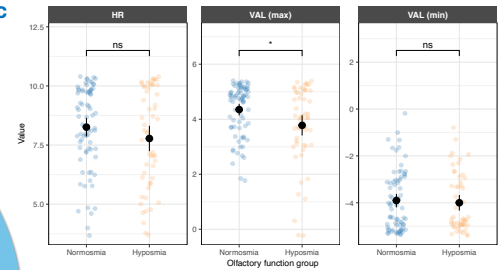


Persistent below-average olfactory function

Hedonic perception

- Linear mixed models showed significantly reduced maximum valence rating in hyposmic patients compared to normosmic
- Minimum valence rating unaffected; hedonic range numerically lower but not significant
- Significant effects for odors garlic (multiple comparison corrected), cloves, and coffee → complex, food related odors

Selective reduction of upper hedonic perception



Prediction

Which PASC-Symptom is the best predictor of olfactory dysfunction (age and gender corrected)?

Methods: linear regression with lasso selection (parametric), random forest importance ranking (non-parametric)

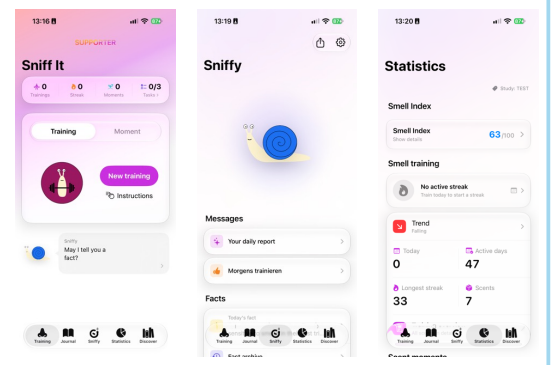
Cognitive function (verbal fluency) as main predictor of olfactory function

Verbal fluency
Respiratory symptoms
Dermatological ailments
Working memory
Parosmia
PHQ-15
Joint/muscle pain
Corticosteroids
Concussion

	LASSO selection (selected)	Linear regression (significant)	Random forest (>1%)
Verbal fluency	X	X	X
Respiratory symptoms	X	X	X
Dermatological ailments	X	X	X
Working memory	X		X
Parosmia	X		X
PHQ-15	X		X
Joint/muscle pain	X		X
Corticosteroids	X		X
Concussion			X

Rehabilitation

Olfactory training is the only validated recovery method but often suffers from low adherence. We developed a digital application for guided and gamified training and journaling.



Long-term olfactory impairment in PASC goes beyond peripheral damage

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Scan for contact, publication & further material



References:

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